Prompt Engineering for Excel Operations

A Beginner's Guide

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1. What is a Prompt?

A **prompt** is an instruction or question that you give to an AI tool (such as ChatGPT, Microsoft Copilot, or Gemini) to perform a specific task.

Example

❌ Bad Prompt

> Calculate sales.

✅ Good Prompt

> Calculate the total sales from the Sales column using the SUM formula and explain each step for a beginner.

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2. Why Learn Prompt Engineering?

Instead of searching for formulas on Google, you can simply ask AI.

Without Prompt Engineering

```
User
↓
Google Search
↓
Read Blogs
↓
Find Formula
↓
Apply Formula
```

With Prompt Engineering

```
User
↓
AI Prompt
↓
Correct Formula
↓
Explanation
↓
Excel Solution
```

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3. Components of a Good Prompt

A good prompt contains five components.

Component	Description	Example
Role	Who should AI act as?	Act as an Excel Tutor
Task	What should AI do?	Calculate total sales
Dataset	Which data to use?	Global Superstore Dataset
Constraints	Any conditions?	Explain for beginners
Output	Desired format	Step-by-step explanation

Prompt Framework

```
Role
↓

Task
↓

Dataset

↓

Constraints

↓

Output
```

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4. Types of Prompts

Type 1

Informational Prompt

Purpose

Learn a concept.

Example

> Explain the SUM function in Excel with simple examples.

Type 2

Formula Generation Prompt

Purpose

Generate Excel formulas.

Example

> Generate an Excel formula to calculate total sales.

Type 3

Formula Explanation Prompt

Purpose

Understand formulas.

Example

> Explain how the VLOOKUP formula works with an example.

Type 4

Data Analysis Prompt

Purpose

Analyze datasets.

Example

> Analyze the Global Superstore dataset and identify the top-selling category.

Type 5

Chart Generation Prompt

Purpose

Suggest charts.

Example

> Suggest the best chart to compare monthly sales.

Type 6

Dashboard Prompt

Purpose

Create dashboards.

Example

> Design an executive dashboard using the sales dataset.

Type 7

Data Cleaning Prompt

Purpose

Clean datasets.

Example

> Identify duplicate records and blank cells.

Type 8

Conditional Formatting Prompt

Purpose

Apply formatting rules.

Example

> Highlight sales greater than ₹50,000.

Type 9

Pivot Table Prompt

Purpose

Generate Pivot Tables.

Example

> Create a Pivot Table showing department-wise expenditure.

Type 10

Automation Prompt

Purpose

Generate VBA or Office Scripts.

Example

> Write VBA code to create a monthly sales report automatically.

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5. Prompt Template

```
Act as an Excel Expert.

Task:
Calculate total sales.

Dataset:
Global Superstore

Constraints:
Explain every step.

Output:
Formula + Explanation + Expected Result
```

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6. Beginner-Level Prompts

Formula Prompts

1.

> Calculate total sales using the SUM formula.

2.

> Calculate average sales using the AVERAGE function.

3.

> Find the highest sales using MAX.

4.

> Find the minimum sales using MIN.

5.

> Count total orders using COUNT.

6.

> Count non-empty cells using COUNTA.

7.

> Calculate total sales for the Furniture category.

8.

> Count orders from the West region.

9.

> Find sales greater than ₹5000.

10.

> Calculate profit percentage.

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7. Intermediate Prompts

Lookup

> Create an XLOOKUP formula to retrieve customer names using Order ID.

FILTER

> Display only Technology category orders.

SORT

> Sort sales in descending order.

UNIQUE

> Display unique customer names.

TEXT

> Extract the month from the Order Date.

IF

> Classify sales into High, Medium and Low.

Nested IF

> Assign grades based on marks.

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8. Advanced Prompts

Pivot Table

> Create a Pivot Table showing Region-wise Sales.

Dashboard

> Build an interactive sales dashboard.

KPI

> Create KPIs for Revenue, Profit and Quantity.

Power Query

> Explain how to merge two Excel sheets using Power Query.

VBA

> Generate VBA code to email the monthly report.

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9. Good vs Bad Prompt

Bad

```
Calculate.
```

Better

```
Calculate total sales.
```

Best

```
Act as an Excel Trainer.

Using the Global Superstore dataset,

calculate total sales from the Sales column using the SUM formula.

Explain every step for beginners.

Provide the formula and expected result.
```

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10. Prompt Improvement Example

Prompt 1

```
Calculate sales.
```

AI Response

❌ Ambiguous

Prompt 2

```
Calculate total sales.
```

AI Response

✔ Better

Prompt 3

```
Using the Global Superstore dataset,

calculate total sales using the SUM function.

Explain each part of the formula.

Show expected output.

Explain for beginners.
```

AI Response

⭐⭐⭐⭐⭐

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11. Prompt Writing Checklist

Before submitting your prompt, ask yourself:

✅ Did I define the role?

✅ Did I specify the task?

✅ Did I mention the dataset?

✅ Did I provide constraints?

✅ Did I specify the output format?

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12. Common Mistakes

❌ Using vague prompts

❌ Missing column names

❌ No dataset information

❌ Asking multiple unrelated questions

❌ Not specifying the output format

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13. AI Prompt Formula (Easy to Remember)

R-T-D-C-O Framework

Letter	Meaning	Example
R	Role	Act as an Excel Trainer
T	Task	Calculate total sales
D	Dataset	Global Superstore 2016
C	Constraints	Explain for beginners
O	Output	Formula + Steps + Result

Template:

```
Act as an Excel Trainer.

Task:
<What should AI do?>

Dataset:
<Name of dataset>

Constraints:
<Any conditions?>
```

```
Output:
<How should the answer be presented?>
```

Suggested Hands-on Activities

1. Write a prompt to calculate total sales.
2. Improve a vague prompt into a detailed prompt using the RTDCO framework.
3. Ask AI to generate an `IF` formula.
4. Ask AI to create a Pivot Table.
5. Ask AI to recommend a chart for sales analysis.
6. Ask AI to clean duplicate records.
7. Ask AI to build a simple dashboard.
8. Compare AI responses from a basic prompt and an improved prompt.

This content is ideal for a **60–90 minute beginner workshop** and provides a clear progression from understanding prompts to applying them effectively for common Excel operations.